Vaccine value profile for cytomegalovirus.

PMID: 37806805 | Indexed in PubMed

Cytomegalovirus (CMV) is the most common infectious cause of congenital malformation and a leading cause of developmental disabilities such as sensorineural hearing loss (SNHL), motor and cognitive deficits. The significant disease burden from congenital CMV infection (cCMV) led the US National Institute of Medicine to rank CMV vaccine development as the highest priority. An average of 6.7/1000 live births are affected by cCMV, but the prevalence varies across and within countries. In contrast to other congenital infections such as rubella and toxoplasmosis, the prevalence of cCMV increases with CMV seroprevalence rates in the population. The true global burden of cCMV disease is likely underestimated because most infected infants (85-90 %) have asymptomatic infection and are not identified. However, about 7-11 % of those with asymptomatic infection will develop SNHL throughout early childhood. Although no licensed CMV vaccine exists, several candidate vaccines are in development, including one currently in phase 3 trials. Licensure of one or more vaccine candidates is feasible within the next five years. Various models of CMV vaccine strategies employing different target populations have shown to provide substantial benefit in reducing cCMV. Although CMV can cause end-organ disease with significant morbidity and mortality in immunocompromised individuals, the focus of this vaccine value profile (VVP) is on preventing or reducing the cCMV disease burden. This CMV VVP provides a high-level, comprehensive assessment of the currently available data to inform the potential public health, economic, and societal value of CMV vaccines. The CMV VVP was developed by a working group of subject matter experts from academia, public health groups, policy organizations, and non-profit organizations. All contributors have extensive expertise on various elements of the CMV VVP and have described the state of knowledge and identified the current gaps. The VVP was developed using only existing and publicly available information.

Congenital cytomegalovirus infection: an overview.

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Cytomegalovirus (CMV) remains the most common cause of congenitally-acquired infection in the United States and is a leading infectious cause of sensorineural hearing loss, cognitive delay, and permanent neurologic sequelae. Although most cases of congenital CMV infection are asymptomatic, significant morbidity and mortality exist for symptomatic infants and may also occur in asymptomatic ones. Diagnosis remains relatively straightforward, but treatment options are limited and associated with some toxicity. Efforts at prevention via vaccination, screening, and improved epidemiology deserve high priority to limit the impact of this common infection.

Update on congenital cytomegalovirus infection: Prenatal prevention, newborn diagnosis, and management.

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Cytomegalovirus (CMV) is the leading cause of congenital infection and the most common cause of non-genetic sensorineural hearing loss (SNHL) in childhood. Although most infected infants are asymptomatic at birth, the risk for SNHL and other neurodevelopmental morbidity makes congenital CMV (cCMV) a disease of significance. Adherence to hygienic measures in pregnancy can reduce risk for maternal CMV infection. The prompt identification of infected infants allows early initiation of surveillance and management. A multidisciplinary approach to management is critical to optimize outcomes in affected infants.

[Congenital cytomegalovirus infection].

PMID: 16205606 | Indexed in PubMed

The cytomegalovirus (CMV) is responsible for the most common congenital infection and represents the most important cause of mental disability and non hereditary sensorineural deafness in infants. One percent to 5% of women show symptoms of infection during pregnancy. Forty thousand newborns are affected by this infection in the United States every year, at a cost of 1.9 billion dollars; in Italy, approximately 5,500 infected newborns are expected each year, including 350 in Lombardia alone. In 90% of the cases, the infected newborns present no symptoms at birth, but, in 10% to 15% of these cases, they are not immune to future manifestations of the infection. Ten percent of newborns with congenital infection, however, show symptoms that lead to a strongly unfavourable prognosis. The authors describe the most important pre and post natal diagnostic findings and emphasize the importance of counseling in this pathology. They also point out the least frequent clinical manifestations of the infection in newborns, which can cause problems of differential diagnosis. In light of the most recent literary data, the article outlines the most recent therapeutic schemes as well as their effectiveness. Lastly, the authors discuss the sequels of the congenital infection, which can present themselves in infected newborns even after a long period of time. This corresponds to one of the most important aspects of the topic and the authors propose a follow-up program that applies to those children until the age of 6. There is currently no effective or safe vaccine for the CMV, therefore hygienic-health measures constitute the principal form of prevention of the CMV congenital infection, which represents a significant problem for newborns, their families, and society.

Congenital cytomegalovirus infections.

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Congenital cytomegalovirus (CMV) infection is one of the most common viral causes of congenital infections in high resource countries and a leading cause of hearing loss as well as an important contributor to neurodevelopmental disabilities in children. During early pregnancy, CMV has a teratogenic potential and may cause malformations such as migrational disturbances in the brain, which can be visualised using neuroimaging methods such as magnetic resonance imaging (MRI) in such children. As a consequence of variation in epidemiology and seropositivity in fertile women, the prevalence of congenital CMV in their offspring varies in different countries between 0.15-2.0%. Some 10-20% of all children with congenital CMV infections exhibit signs of neurological damage when followed up. This is the case in children both with and without symptoms of infection at birth. Until vaccines and non-toxic antiviral agents are available, hygienic measures are important as prophylaxis. Treatment with antiviral agents may have a place in children with central nervous system involvement during the neonatal period.

Congenital cytomegalovirus - who, when, what-with and why to treat?

PMID: 28646968 | Indexed in PubMed

Congenital cytomegalovirus (CMV) is the commonest cause of congenital infection worldwide and the leading non-genetic cause of sensorineural hearing loss in children. Appropriate investigations and timely decision on treatment is required as studies have shown that treatment with antiviral therapy leads to improved hearing and neurodevelopmental outcomes in the long term when started in the first month of life. This paper outlines the epidemiology, investigations in the diagnosis of congenital CMV infection and current evidence surrounding treatment.

Late-onset and congenital hearing loss detected using AABR due to congenital cytomegalovirus infection that improved with valganciclovir.

PMID: 36517460 | Indexed in PubMed

Congenital cytomegalovirus (cCMV) infection is the most common congenital viral infection and is the leading non-genetic cause of sensorineural hearing loss and an important cause of neurodevelopmental disabilities. Auto auditory brainstem response (AABR) is a simple hearing test and used for the purpose of neonatal hearing screening, but can use it for early detection hard of hearing within the study age of the model. We experienced two case of asymptomatic CMV infection in which congenital and late-onset hearing loss were diagnosed early with AABR, and hearing loss improved with valganciclovir.

Induction of cytomegalovirus-infected labyrinthitis in newborn mice by lipopolysaccharide: a model for hearing loss in congenital CMV infection.

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Congenital cytomegalovirus (CMV) infection is the most common infectious cause of sensorineural hearing loss in children. Here, we established an experimental model of hearing loss after systemic infection with murine CMV (MCMV) in newborn mice. Although almost no viral infection was observed in the inner ears and brains by intraperitoneal (i.p.) infection with MCMV in newborn mice, infection in these regions was induced in combination with intracerebral (i.c.) injection of bacterial lipopolysaccharide (LPS). The susceptibility of the inner ears was higher than that of the brains in terms of viral titer per unit weight. In the labyrinths, the viral infection was associated with the mesenchymal vessels and accompanied by inflammatory cells induced by LPS, causing hematogenous targets of infection in the labyrinths. Viral infection also spread in the perilymph regions such as the scala tympani and scala vestibuli, probably from infected brains via meningogenic and cochlear nerve routes. Viral infection was not observed in the scala media in the endolymph, including the Corti organ. However, viral infection was observed in the spiral limbus, including the stria vascularis. These results suggest that hearing loss caused by labyrinthitis after congenital CMV infection may be enhanced by inflammation caused by systemic bacterial infection in the neonatal period.

Congenital cytomegalovirus labyrinthitis and sensorineural hearing loss in guinea pigs.

PMID: 2555420 | Indexed in PubMed

Cytomegalovirus (CMV) is the leading cause of human nonhereditary congenital deafness. The pathogenesis of congenital CMV infection in the auditory system is poorly understood and no suitable animal model is currently recognized. In this study primary maternal CMV infection in guinea pigs during the first or second trimester of pregnancy resulted in congenital infection in 64% of the offspring. Of the congenitally infected neonates, 28% had significant auditory deficits. Within the inner ear, CMV infection was localized in auditory nerve spiral ganglion cells. These findings indicate that congenital CMV infection of the guinea pig results in physiologic and anatomic neuropathology similar to that seen in human infection and provide the first experimental model for congenital CMV-induced sensorineural hearing loss.

Guinea pig model of transplacental congenital cytomegaloviral infection with analysis for labyrinthitis.

PMID: 1647141 | Indexed in PubMed

Cytomegalovirus is the most frequently recognized cause of congenital viral infection and of viral induced congenital hearing loss. Histopathologic reports on temporal bones from nine congenitally infected infants have most often demonstrated an endolabyrinthitis. The guinea pig and its specific cytomegaloviruses have been studied extensively as a model of transplacentally acquired infection. However, to date no investigation has been performed on the inner ear of such congenitally infected animals. To determine if a congenital viral labyrinthitis occurs in these guinea pigs, a study was conducted in which pregnant Hartley and Strain 2 guinea pigs were injected intraperitoneally with virulent salivary gland passaged cytomegalovirus during the first, second, or third trimester. Various durations of gestation up to delivery were permitted and selected organs from the products were then studied with routine histologic and immunocytochemical methods and cell cultures to verify the presence of transplacental infection. The temporal bones were studied with routine histopathology as well as immunocytochemical methods to detect cytomegalovirus antigen. Over 190 products of conception studied in this fashion revealed no evidence of teratogenesis or labyrinthine infection. It is not clear why in the presence of viremia, which resulted in cytomegalovirus infection of multiple organs, the inner ears of these animals were apparently spared.
